CAPSTONE PROJECT REPORT · CAUSAL MACHINE LEARNING

Who Will Respond?

Uplift Modeling for Targeted Offers

Predicting who responds because of an offer — and what it is worth

Lokesh Maan · October 2026 · github.com/LokeshMaan10/uplift

2.1×	**−45%**	**7/7**
campaign profit vs mailing everyone	fewer offers sent	uplift models beat every baseline

Abstract. Companies send offers to entire customer bases, although most customers either respond anyway or never will. This project builds uplift models that estimate, for each customer, the *causal* effect of an offer on the probability of responding, using a randomized e-mail experiment with 42,693 customers. Seven uplift estimators (S-, T-, X- and R-learners on LightGBM, class transformation, uplift random forest and causal forest) are compared with response-model baselines using Qini and uplift curves, cross-validation and bootstrap confidence intervals. All uplift models clearly beat the baselines; the standard response model is statistically indistinguishable from random targeting (Qini 0.011 vs 0.010). Translated into money, uplift targeting earns $73.8k instead of $35.3k per 100,000 customers while contacting only 55% of them. Results are explained with SHAP, stress-tested across random seeds, a second treatment arm and 500,000 Criteo users, and delivered as an interactive Streamlit dashboard.

Stack: Python 3.11 · pandas · scikit-learn · LightGBM · scikit-uplift · CausalML · EconML · SHAP · MLflow · Streamlit · pytest. 100% free tools and public data, no API keys.

Contents

Executive summary 3

1. Business problem 4

2. Data and experiment validation 4

3. Causal concepts 6

4. Methodology and architecture 7

5. Results 8

6. Business impact 9

7. Explainability and segments 11

8. Robustness 12

9. Engineering and deployment 14

10. Limitations and future work 14

11. Conclusion 15

Executive summary

Problem. A response model ranks customers by how likely they are to buy. That puts customers who would buy anyway ("Sure Things") at the top of the mailing list, so the campaign pays for sales it did not cause. The business question is different: *whose behaviour does the offer change?* That quantity — the individual treatment effect, or **uplift** — is what this project estimates.

Approach. Using the randomized Hillstrom e-mail experiment, I first verified that treatment was truly random (propensity AUC 0.497), measured the average effect (+4.5 pp), and showed that it varies strongly between customers. I then trained and compared ten models — three baselines and seven uplift estimators — under one interface, evaluated them on a held-out test set with uplift-specific metrics, converted the rankings into campaign profit, explained them with SHAP, and checked robustness on new splits and new data.

Key findings

- **Every uplift model beats every baseline.** Test Qini coefficients range from 0.047 to 0.072, all with 95% confidence intervals above zero. The response model scores 0.011 — the same as random (0.010).
- **About twice the profit with half the mail.** At $1 per offer and $30 per extra response, targeting the top 55% by uplift yields $73.8k versus $35.3k for mailing everyone (+$38.5k per 100,000 customers).
- **The model knows when to stop.** As offers get more expensive the optimal target group shrinks; from $3 per offer the best decision is not to run the campaign.
- **Who is persuadable:** past buyers of women's products (the treatment was the women's e-mail). Customers the model labels Persuadable showed an observed uplift of +8.2 pp; Sure Things showed +0.6 pp.
- **Honest limits.** Uplift modeling only pays when effects differ between customers and are not aligned with baseline response: on the men's e-mail no model beats random, and on Criteo a response model performs as well as the uplift models.

> **Recommendation.** Replace response-model targeting with an uplift model, choose the share of customers to contact from the profit curve for the current offer cost, and keep a small random control group in every campaign so uplift can be re-measured and the model retrained.

1. Business problem

A bank sending credit-card or loan offers — or a retailer sending promotional e-mails — usually contacts everyone, or everyone a response model rates highly. Both waste money, because the decision that matters is not whether a customer will respond, but whether the offer *changes* that response. Every customer falls into one of four groups, defined by what they would do with and without the offer:

Segment	Without offer	With offer	Effect of the offer	Action
Persuadable	does not respond	responds	creates a response	**Target**
Sure Thing	responds	responds	none — cost is wasted	Skip
Lost Cause	does not respond	does not respond	none — cost is wasted	Skip
Sleeping Dog	responds	does not respond	destroys a response	**Never target**

An uplift model estimates, for a customer with features x, the conditional average treatment effect (CATE):

> $\tau(x) = P(\text{respond} \mid \text{offer}, x) - P(\text{respond} \mid \text{no offer}, x)$

Ranking customers by $\tau(x)$ puts Persuadables first and Sleeping Dogs last. The project goal is to build that ranking, prove it is better than the alternatives, and translate it into money saved.

2. Data and experiment validation

All data is public and loaded with free tools. The main dataset is the Hillstrom MineThatData e-mail experiment (via scikit-uplift): 64,000 customers were randomly assigned to receive a men's e-mail, a women's e-mail, or no e-mail. The outcome is a website visit within two weeks. Purchases (~1%) are too rare to model reliably, so visits are used as the response.

Dataset	Rows used	Treatment	Outcome	Role
Hillstrom (women's e-mail vs none)	42,693	e-mail sent (50%)	visit	main analysis
Hillstrom (men's e-mail vs none)	42,613	e-mail sent (50%)	visit	robustness
Criteo Uplift v2.1	500,000 of 14M	ad shown (85%)	visit	scale + 2nd domain

Features: months since last purchase, spend in the past year, whether the customer bought men's and/or women's merchandise, new-customer flag, area (urban / suburban / rural) and purchase channel. The binned spend column was dropped as a duplicate of spend. The scikit-uplift Criteo download links no longer work, so the loader fetches Criteo's official file from its public Hugging Face repository and samples it in chunks to stay within laptop memory.

2.1 Was treatment really random?

Uplift estimates are only causal if the treated and control groups were comparable before the e-mail. Three checks confirm this:

- **Standardized mean differences:** the largest across all 11 features is 0.009, far below the usual 0.1 threshold.
- **t-tests:** 0 of 11 features differ at $p < 0.05$ (about 0.6 expected by chance).

- **Propensity AUC = 0.497:** a model trying to predict who received the e-mail from customer features does no better than a coin flip.

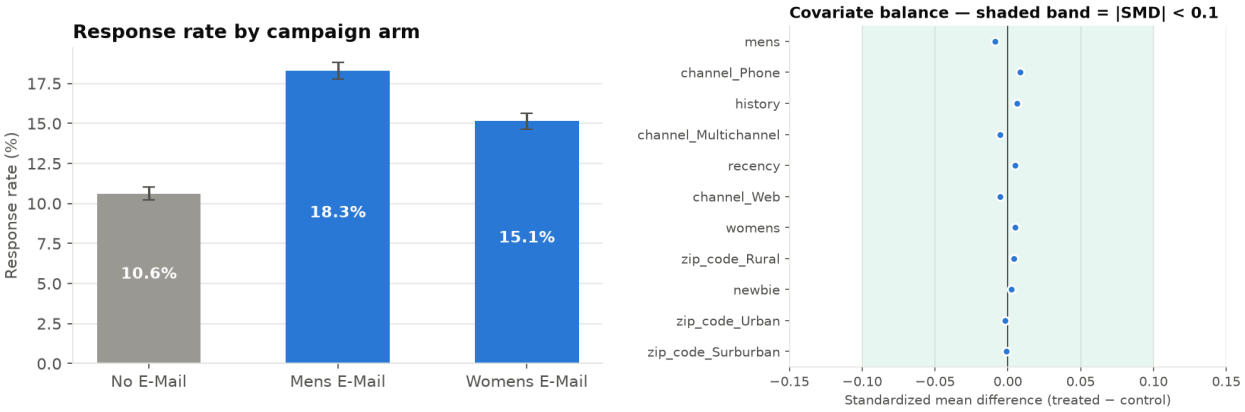


Figure 1. Left: visit rate by campaign arm. Right: covariate balance — every feature lies inside the ±0.1 band.

2.2 The average effect hides large differences

The women's e-mail lifted visits from 10.6% to 15.1%: an average treatment effect of **+4.52 pp** (95% CI +3.89 pp to +5.16 pp), a 43% relative lift. The men's e-mail had an even larger average effect (+7.7 pp). But the effect is not uniform: among past buyers of women's products the women's e-mail adds about +7 pp, among men's-only buyers about +1 pp. A single average would waste most of the budget — this heterogeneity is exactly what uplift models exploit.

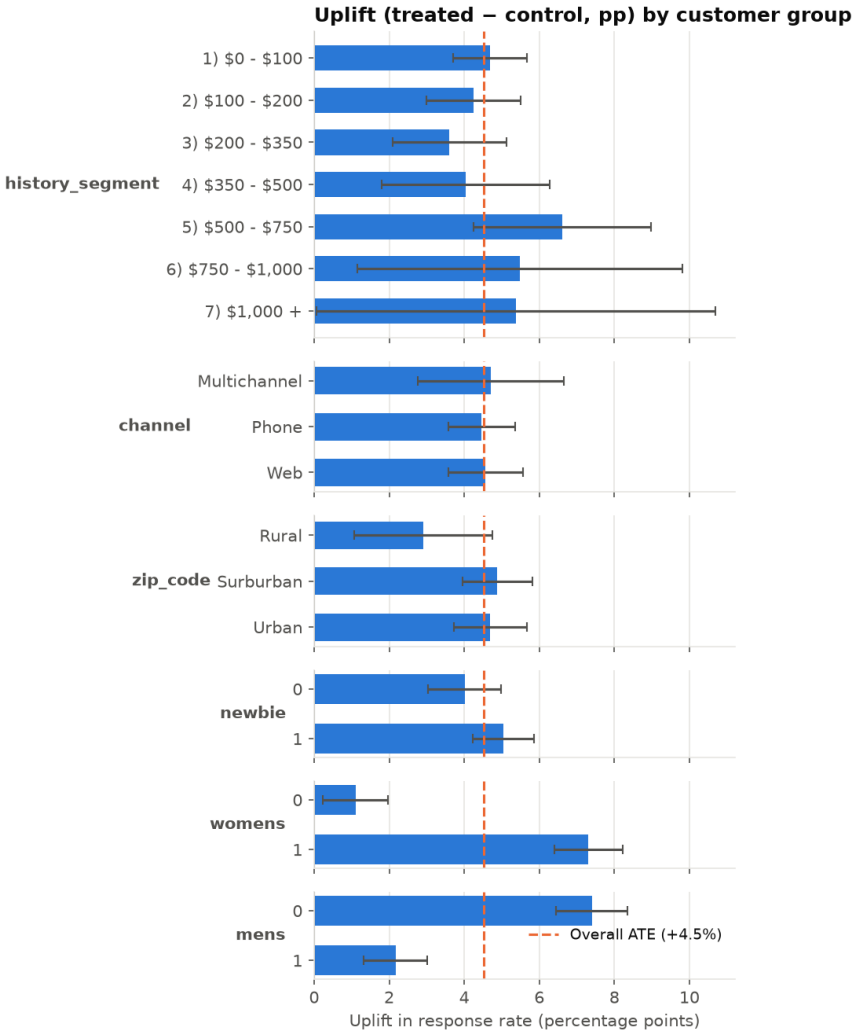


Figure 2. Uplift (treated − control) within customer groups, with 95% CIs. Dashed line = overall ATE.

3. Causal concepts

Potential outcomes. Each customer has two possible futures: Y(1), what they do if contacted, and Y(0), what they do if not. The individual effect is Y(1) − Y(0). **The fundamental problem of causal inference** is that only one of the two is ever observed, so individual effects can never be measured directly. Instead we estimate the average effect for customers with similar features — the CATE, τ(x).

ATE vs CATE. The average treatment effect (ATE) is one number for everyone and, with randomization, equals the difference in response rates between the groups. The CATE varies with customer features and is what decides whom to target.

To make these ideas visible, I simulated a bank where both potential outcomes are known. In the simulation, 10% of customers are Persuadables, 23% Sure Things, 63% Lost Causes and 3% Sleeping Dogs, with a true ATE of +7.0 pp. Giving three targeting strategies a perfect model of their own target and the same budget (top 20%) shows why the question matters:

Strategy (perfect model, top 20%)	Extra responses per 1,000 contacted	Sleeping Dogs contacted
Historical-buyer model	-90	14.5%
Campaign response model	+136	1.4%
Uplift model	+292	0.0%

The classic "propensity to buy" model *loses* responses because loyal, engaged customers — where Sleeping Dogs live — look like the best prospects. A response model trained on campaign data fills its budget with Sure Things. The uplift ranking roughly doubles the incremental responses.

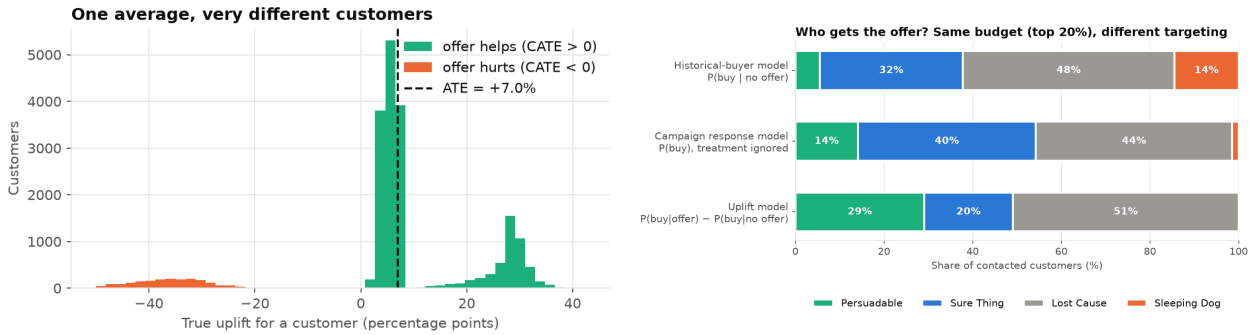


Figure 3. Simulated bank. Left: one ATE, very different individual effects. Right: who each strategy contacts.

Assumptions. (1) Unconfoundedness — treatment is independent of potential outcomes given x, guaranteed here by randomization; (2) overlap — every customer type can appear in both groups; (3) SUTVA — one customer's treatment does not affect another's outcome.

4. Methodology and architecture

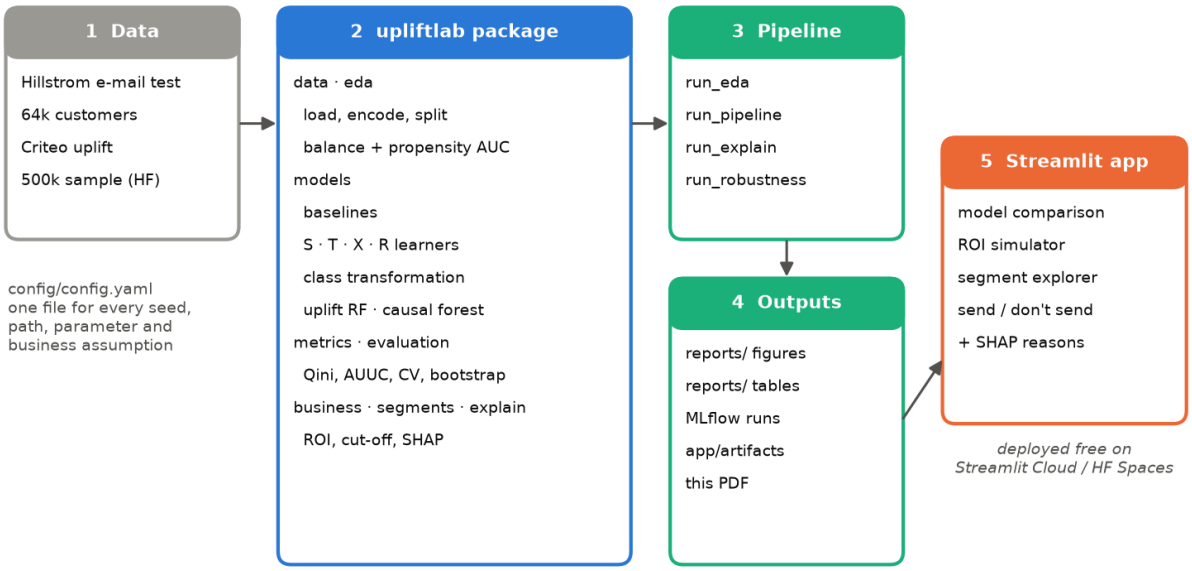


Figure 4. System architecture: one package, four pipeline scripts, shared outputs, and a lightweight dashboard.

4.1 Models

Every model implements the same two methods — **fit(X, y, t)** and **predict_uplift(X)** — so evaluation, ROI and the app treat them identically. The meta-learners are implemented from scratch on LightGBM.

Model	How it estimates uplift	Strength / weakness
Random	random score	reference: what untargeted mailing achieves
Response model	P(respond | x), treatment ignored	standard industry practice; wrong question
Historical buyer	P(respond | x, no offer)	"propensity to buy"; targets Sure Things
S-learner	one model f(x, t); f(x,1) − f(x,0)	simple; can under-use t (shrinks uplift)
T-learner	μ1(x) − μ0(x), one model per group	flexible; noise of two models adds up
X-learner	impute effects with the other group's model, then model them; propensity-weighted	good for unbalanced groups
R-learner	Robinson residuals: (y − m(x)) on (t − e), cross-fitted	targets the CATE directly; robust to outcome-model error
Class transformation	Z = 1 if (treated & responded) or (control & not); uplift = 2P(Z) − 1	one classifier; reweighted for unbalanced splits
Uplift random forest	trees split on treated-vs-control divergence (KL), causalml	purpose-built; slowest
Causal forest	double ML + honest forest on residuals, econml	statistically grounded; stable

4.2 Evaluation protocol

- 70/30 train/test split, stratified on treatment × outcome so both parts keep the same treated share and response rate.

- 5-fold stratified cross-validation on the training set for model comparison and tuning; the test set is used once for final numbers.
- Metrics: **Qini coefficient** (area between the model's Qini curve and random, normalised by a perfect model; 0 = random, 1 = perfect), **AUUC**, **uplift@k%** (treated − control response rate in the top k%), and uplift by decile.
- **Bootstrap** (500 resamples of the test set) for 95% confidence intervals on the Qini coefficient.
- A known-truth check on simulated data, where each model's predictions are compared with the true uplift.

4.3 Hyper-parameters chosen without touching the test set

The uplift signal in Hillstrom is weak, so flexible trees fit noise. Three shared LightGBM settings were compared by cross-validated Qini on training data only; strong regularisation won and was used for all LightGBM-based models.

Setting	Mean CV Qini (5 learners)
flexible (300 trees, 31 leaves)	0.033
regularized (200 trees, 15 leaves)	0.041
strong (150 trees, 7 leaves)	0.048

5. Results

Table 5.1 compares all ten models on the held-out test set of 12,808 customers.

Model	Family	Qini (test)	95% CI	CV Qini	Uplift top 10%	Uplift top 30%
Uplift RF (KL)	Other uplift	0.072	[0.045, 0.099]	0.052 ± 0.020	+5.0 pp	+9.2 pp
S-learner	Meta-learner	0.069	[0.041, 0.097]	0.046 ± 0.017	+6.6 pp	+8.2 pp
Causal forest	Other uplift	0.066	[0.037, 0.093]	0.062 ± 0.015	+6.3 pp	+8.2 pp
X-learner	Meta-learner	0.064	[0.035, 0.091]	0.053 ± 0.018	+3.4 pp	+7.8 pp
R-learner	Meta-learner	0.058	[0.029, 0.087]	0.048 ± 0.019	+5.3 pp	+8.4 pp
T-learner	Meta-learner	0.058	[0.028, 0.086]	0.049 ± 0.016	+4.3 pp	+6.8 pp
Class transformation	Other uplift	0.047	[0.017, 0.073]	0.042 ± 0.014	+5.8 pp	+7.1 pp
Response model	Baseline	0.011	[-0.012, 0.039]	0.023 ± 0.021	+6.3 pp	+6.3 pp
Random	Baseline	0.010	[-0.016, 0.041]	0.011 ± 0.020	+5.1 pp	+6.1 pp
Historical buyer	Baseline	-0.011	[-0.038, 0.017]	0.005 ± 0.023	+6.1 pp	+3.9 pp

- **All seven uplift models beat all three baselines**, and all their confidence intervals exclude zero. The response model's interval includes zero: it is no better than random targeting.
- The top four models' intervals overlap, so they are statistically tied. Uplift RF (KL) had the highest test Qini; the causal forest had the best and most stable cross-validated score. The dashboard uses the S-learner — pure LightGBM, fast and small to deploy.
- Uplift@10% is similar for all models (the very top slice is noisy); the difference shows up across the first 30–50% of customers, which is where campaign profit is decided.
- **Known-truth check:** on simulated data, uplift models reached Qini 0.11–0.12 against a perfect 0.15, while the response model reached only 0.03.

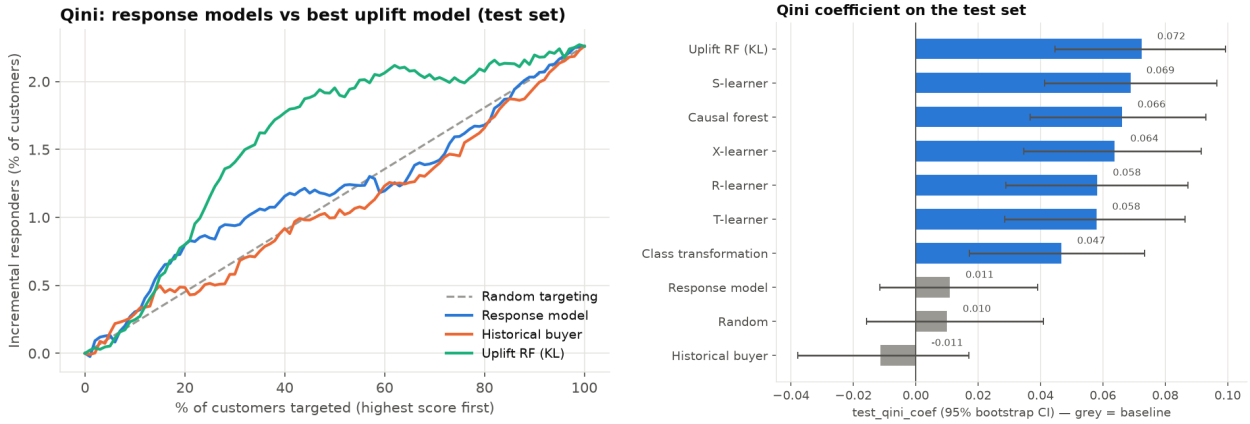


Figure 5. Left: Qini curves — the uplift model gathers incremental responders early, response models follow the random line. Right: test Qini with bootstrap 95% CIs.

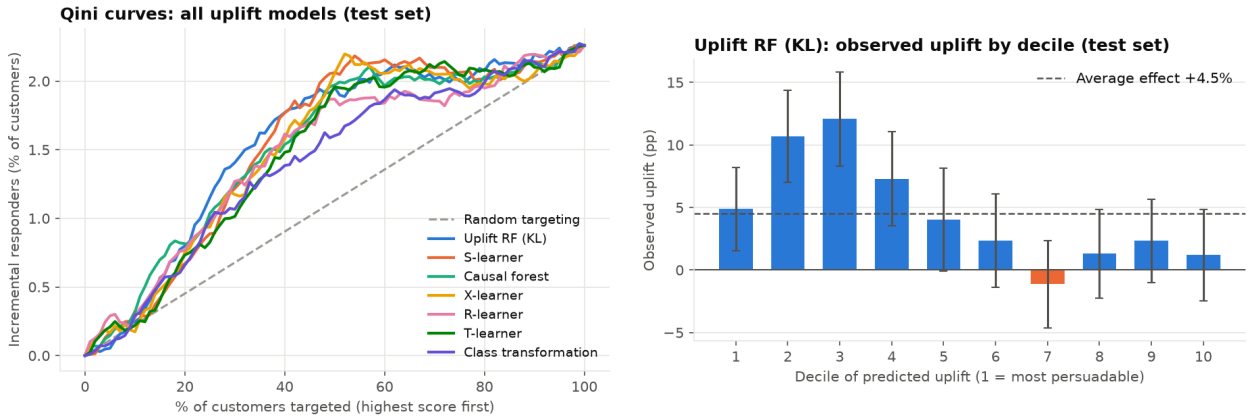


Figure 6. Left: Qini curves for all uplift models. Right: observed uplift by decile for the Uplift RF (KL).

6. Business impact

For each strategy and each targeted share p, the uplift among the top p% is measured on the randomized test set and scaled to a campaign of 100,000 customers, with **$1 per offer** and **$30 profit per extra response**. Because uplift is measured rather than predicted, the profit figures are causal estimates.

> profit(p) = uplift(top p%) × customers targeted × value − customers targeted × cost

Strategy	Best % to target	Profit	ROI	Extra responses
S-learner	55%	$73.8k	134%	4,295
Uplift RF (KL)	50%	$66.3k	133%	3,878
Response model	100%	$35.3k	35%	4,511
Historical buyer	100%	$35.3k	35%	4,511
Mail everyone	100%	$35.3k	35%	4,511

Uplift targeting earns **+$38.5k** more than mailing everyone — 2.1× the profit — while contacting only 55% of customers. Neither response-style model can beat mailing everyone: their best option is to contact all customers.

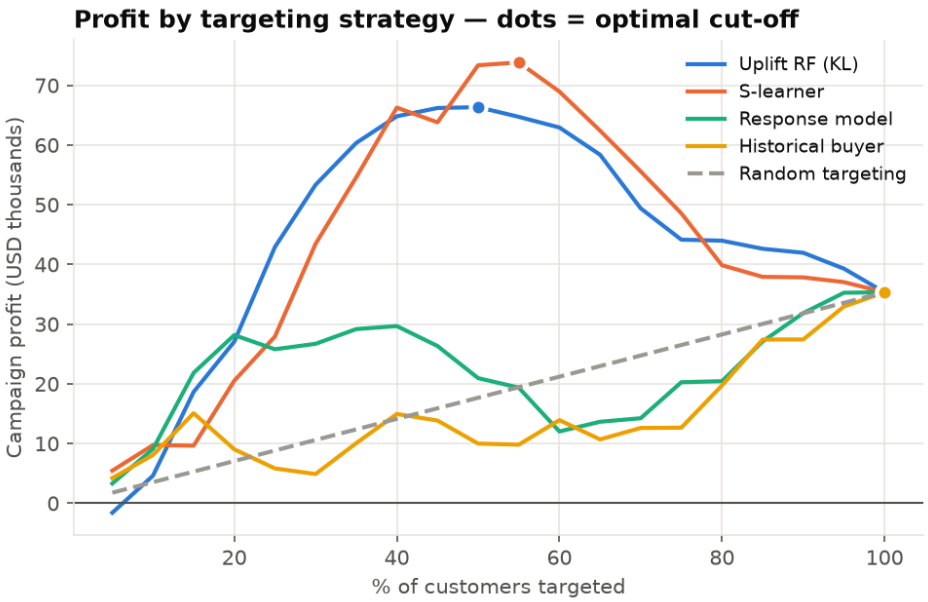


Figure 7. Profit by share of customers targeted; dots mark each strategy's optimum.

6.1 What-if: offer cost

The optimal plan depends on the offer cost. A customer is worth contacting only if their uplift exceeds the break-even rate cost ÷ value. Table 6.2 repeats the analysis for the Uplift RF (KL).

Cost per offer	Break-even uplift	Optimal % to target	Profit (model)	Profit (mail all)
$0.25	0.8%	95%	$110.5k	$110.3k
$0.50	1.7%	60%	$92.9k	$85.3k
$1.00	3.3%	50%	$66.3k	$35.3k
$1.50	5.0%	40%	$44.8k	−$14.7k
$2.00	6.7%	35%	$25.3k	−$64.7k
$3.00	10.0%	don't run	$0	−$164.7k
$4.00	13.3%	don't run	$0	−$264.7k
$5.00	16.7%	don't run	$0	−$364.7k

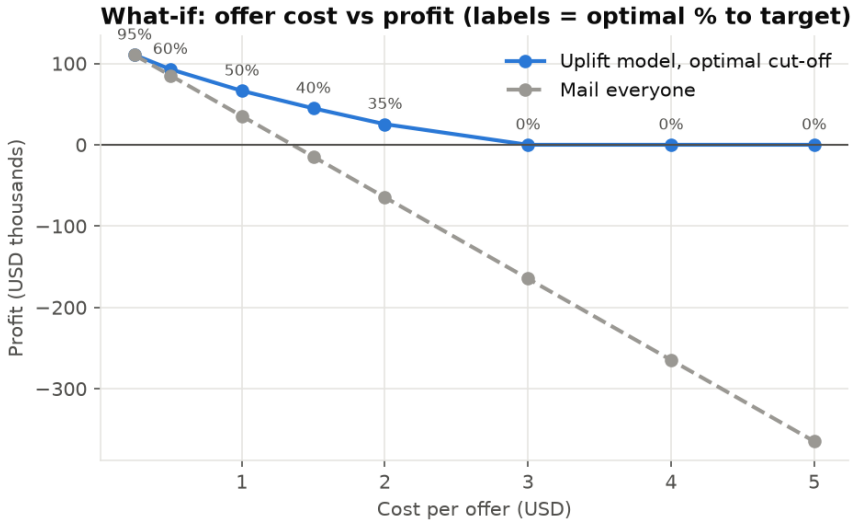


Figure 8. As offers get more expensive the model narrows targeting; mailing everyone turns loss-making much sooner.

7. Explainability and segments

Meta-learners combine several sub-models, so there is no single tree to inspect. A surrogate LightGBM model was trained to mimic each model's predicted uplift (fidelity R^2 0.98 for the S-learner) and explained with SHAP. The direction column shows whether a higher feature value increases (+) or decreases (−) predicted uplift.

Feature	Mean |SHAP| (pp)	Direction
womens	2.31	+
history	0.20	−
zip_code_Rural	0.13	−
recency	0.12	+
newbie	0.11	−
mens	0.10	+

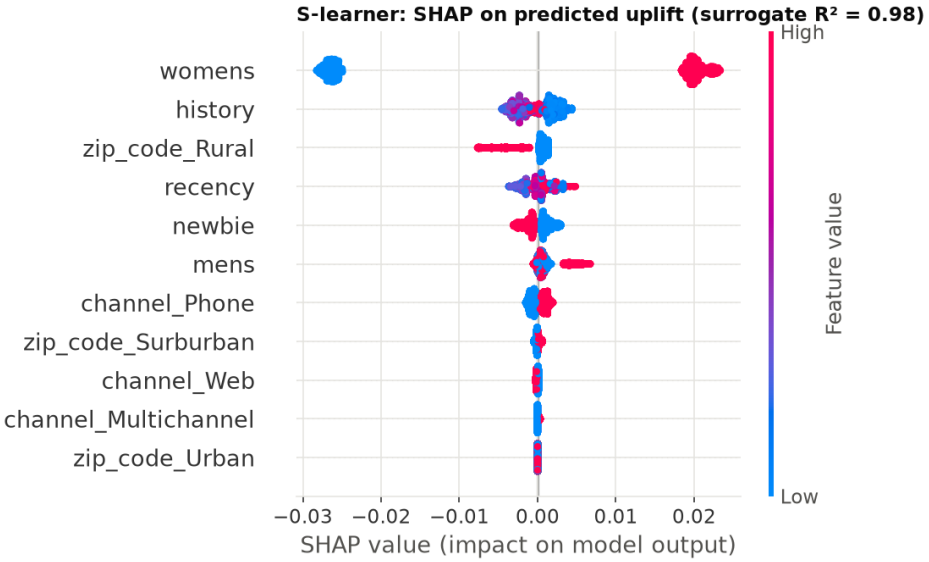


Figure 9. SHAP values on predicted uplift (S-learner surrogate).

Interpretation: the e-mail promoted women's merchandise, so customers who bought women's products last year are by far the most persuadable; men's-only buyers and rural customers react least; high spenders are slightly less persuadable because they tend to visit anyway.

7.1 Do predicted segments behave as labelled?

Customers were labelled from predictions: uplift below 0 → Sleeping Dog; top 30% of uplift → Persuadable; otherwise Sure Thing if their baseline response is above the median, else Lost Cause. Because the test set is randomized, each segment's true uplift can be measured independently:

Segment	Share	Predicted uplift	Observed uplift (95% CI)	Baseline rate
Persuadable	30%	+6.7 pp	+8.2 pp [+6.0 pp, +10.5 pp]	11.2%
Sure Thing	31%	+2.8 pp	+0.6 pp [-1.7 pp, +3.0 pp]	16.7%
Lost Cause	37%	+2.9 pp	+4.9 pp [+3.4 pp, +6.4 pp]	5.0%
Sleeping Dog	2%	-0.5 pp	+0.5 pp [-8.3 pp, +9.3 pp]	14.5%

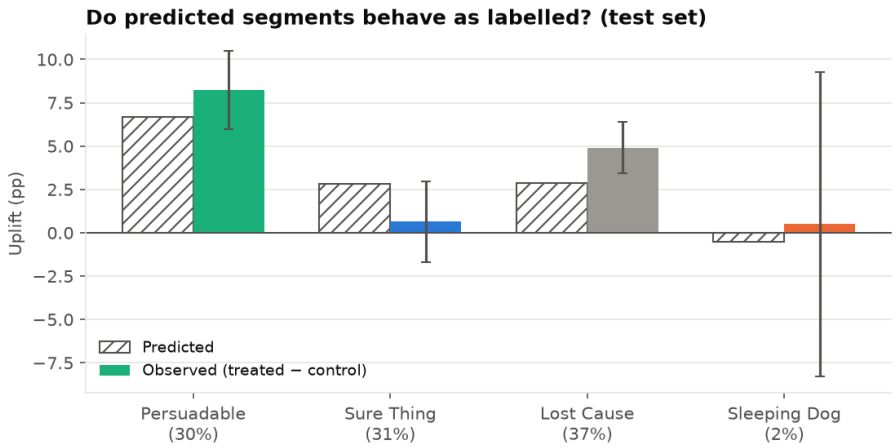


Figure 10. Predicted vs observed uplift per segment.

Persuadables were validated: observed uplift +8.2 pp, almost twice the average. Sure Things were validated too: high baseline (16.7%) and essentially zero uplift. Persuadables are all past women's buyers (100%) and spend $285 on average. The Lost Cause label proved too pessimistic (their uplift is close to average), and the small Sleeping Dog group cannot be confirmed — both are reported as limitations.

8. Robustness

8.1 Stability across random splits

The full split-train-evaluate cycle was repeated with five different seeds on Hillstrom (the slow uplift random forest was excluded). The worst run of any uplift model (Qini 0.041) still beat the response model's average (0.013).

Model	Mean Qini	Std	Worst seed	Mean rank
S-learner	0.062	0.014	0.046	2.6
Causal forest	0.062	0.009	0.053	2.0
X-learner	0.060	0.013	0.050	3.6
T-learner	0.060	0.013	0.047	3.4
Class transformation	0.055	0.009	0.043	4.4
R-learner	0.053	0.011	0.041	5.0
Response model	0.013	0.014	-0.007	7.0
Random	-0.009	0.007	-0.017	8.0

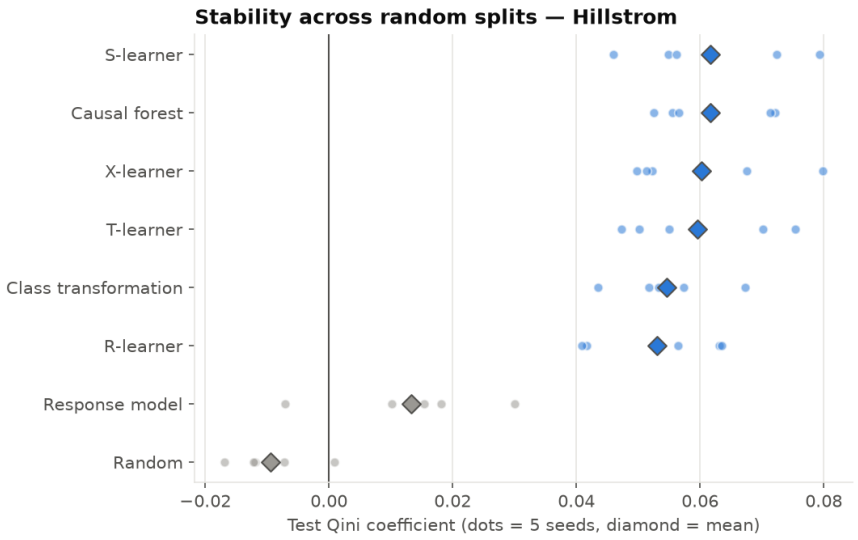


Figure 11. Test Qini for each of five seeds.

8.2 A second treatment: the men's e-mail

The men's e-mail has a larger average effect, but no model beats random targeting: the best Qini is 0.013 and every confidence interval includes zero. The effect is roughly the same for everyone, so there is nothing to target — mailing all customers is the right decision for that campaign. Uplift modeling only adds value when effects are heterogeneous.

8.3 Scale and a second domain: Criteo

On 500,000 Criteo users (85% treated, visit rate about 4.7%, ATE +1.17 pp), all models beat random clearly. However, the response model ties the best uplift models: in this dataset, users likely to visit are also the ones the ads move, so ranking by response probability and by uplift agree.

Model	Qini	95% CI	Uplift@10%
S-learner	0.113	[0.085, 0.139]	+8.0 pp
Response model	0.111	[0.083, 0.139]	+8.0 pp
R-learner	0.105	[0.078, 0.128]	+8.0 pp
X-learner	0.094	[0.066, 0.123]	+8.0 pp
T-learner	0.091	[0.061, 0.121]	+7.1 pp
Class transformation	0.067	[0.042, 0.091]	+4.9 pp
Random	-0.007	[-0.027, 0.010]	+1.1 pp

Takeaway. Uplift models were never worse than the response model in any setting, and far better when uplift and baseline response diverge (Hillstrom). They are the safe default — but checking for heterogeneity first avoids over-engineering campaigns whose effect is uniform.

9. Engineering and deployment

9.1 Interactive dashboard

Tab	Purpose
Model comparison	Qini / uplift curves for any models, results table, uplift by decile
ROI simulator	sliders for offer cost, profit per response and % targeted; profit, ROI and optimal cut-off for each strategy; offer-cost what-if
Segments	segment sizes, predicted vs observed uplift, profiles, filters by channel and area
Should we send the offer?	describe one customer → probabilities with and without the offer, uplift, segment, send / don't send with expected profit and SHAP reasons

9.2 Reproducibility and quality

- **Single configuration file** (config/config.yaml) for every seed, path, hyper-parameter and business assumption.
- **MLflow** logs every model run with parameters, metrics and the fitted model, plus all robustness runs.
- **Automated tests** cover the ATE and balance statistics, every model (including recovery of the true uplift on simulated data), the uplift metrics, the ROI arithmetic and decision rules, the segment rules, SHAP, and a headless run of the whole dashboard.
- **Deployment check:** the app was run in a fresh environment containing only its slim requirements, which exposed and fixed two hidden dependency problems before deployment.
- **Free hosting:** Streamlit Community Cloud (slim requirements file) or Hugging Face Spaces (Dockerfile included). Code: github.com/LokeshMaan10/uplift.

10. Limitations and future work

Limitations

- **Weak signal.** Hillstrom has 11 features and a modest effect, so Qini values are small (~0.07) and the best uplift models are statistically tied.
- **Proxy outcome.** Website visits stand in for offer acceptance; purchases are too rare to model.
- **Assumed economics.** The $1 cost and $30 value are illustrative; the what-if analysis and the dashboard show how conclusions change.
- **Segments are estimates.** True segments are unobservable; the Lost Cause and Sleeping Dog labels did not validate cleanly.
- **Noise in small slices.** Uplift in the top 5–10% rests on a few hundred customers per group.
- **Single campaign.** No drift, seasonality or repeated-contact fatigue can be studied.

Future work

- Model profit directly (spend uplift) instead of visit uplift, and optimise for expected profit per customer.
- Multi-treatment uplift: choose the best of several offers (men's vs women's e-mail) per customer.
- Run each campaign with a random holdout group to monitor real uplift and retrain on fresh data.
- Policy evaluation with doubly robust estimators and budget-constrained targeting.

11. Conclusion

Asking "who will respond?" leads to targeting customers who would respond anyway. Asking "who will respond *because* of the offer?" — and answering it with causal machine learning on randomized data — lets a company spend its marketing budget only where it changes behaviour. In this project, all seven uplift estimators outperformed response-based targeting, the best plans roughly doubled campaign profit while halving the number of offers, and the robustness analysis clarified when the approach pays off and when it does not. The full pipeline — validation, modeling, evaluation, business simulation, explanation and an interactive tool — is reproducible end to end with free software and public data.

Appendix: glossary

Term	Meaning
Uplift / CATE	change in response probability caused by the offer for customers with features x
ATE	average treatment effect over everyone: E[Y(1) − Y(0)]
Qini curve	incremental responders gained as more customers are targeted, best-scored first
Qini coefficient	area between the Qini curve and random, normalised by a perfect model
AUUC	area under the uplift curve, normalised the same way
uplift@k%	treated minus control response rate among the top k% of customers
Propensity AUC	how well features predict treatment; 0.5 means assignment was random
SMD	standardized mean difference between groups; below 0.1 counts as balanced
Bootstrap CI	range of a metric over resampled test sets; shows its uncertainty